

Dynamic Bonding Curve

0.1.9

Smart Contract Security Assessment

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1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers*, *operating systems*, *IoT devices*, and *hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple*, *Google*, and *Microsoft*, have protected digital assets valued at over **\$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **\$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.



<https://offside.io/>



<https://github.com/offsidelabs>



https://twitter.com/offside_labs



2 Executive Summary

Introduction

Offside Labs completed a security audit of *Dynamic Bonding Curve* smart contracts, starting on January 19th, 2026, and concluding on January 23th, 2026.

Project Overview

The Dynamic Bonding Curve (DBC) is a permissionless launch pool protocol that enables users to create and launch tokens directly with customizable bonding curves. The program also allows partners to configure key parameters such as quote tokens, token graduation curves, and fees.

Release 0.1.9 refines DAMM v2 migration parameters and validation, adds protocol migration fee tracking and claiming, strengthens first swap min fee and end of curve leftover handling, and updates virtual pool and config fields.

Audit Scope

The assessment scope contains mainly the smart contracts of the dynamic-bonding-curve program for the *Dynamic Bonding Curve* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Dynamic Bonding Curve
 - Codebase: <https://github.com/MeteoraAg/dynamic-bonding-curve>
 - Release 0.1.9: [PR-165](#)
 - Commit Hash: 437db38cded0fe80fb6ba37b032c839b4fcb93bd

The issues described in this report have been resolved by the following commit:

- Dynamic Bonding Curve
 - Codebase: <https://github.com/MeteoraAg/dynamic-bonding-curve>
 - Commit Hash: b17ce1eed762e198f0d390c5b800d6cfff379dea

We listed the files we have audited below:

- Dynamic Bonding Curve Release 0.1.9
 - programs/dynamic-bonding-curve/src/base_fee/fee_rate_limiter.rs
 - programs/dynamic-bonding-curve/src/base_fee/fee_scheduler.rs
 - programs/dynamic-bonding-curve/src/base_fee/mod.rs
 - programs/dynamic-bonding-curve/src/constants.rs
 - programs/dynamic-bonding-curve/src/error.rs
 - programs/dynamic-bonding-curve/src/event.rs
 - programs/dynamic-bonding-curve/src/instructions/admin/auth.rs
 - programs/dynamic-bonding-curve/src/instructions/creator/ix_claim_creator_trading_fee.rs



- programs/dynamic-bonding-curve/src/instructions/creator/ix_withdraw_creator_surplus.rs
- programs/dynamic-bonding-curve/src/instructions/initialize_pool/ix_initialize_virtual_pool_with_spl_token.rs
- programs/dynamic-bonding-curve/src/instructions/initialize_pool/ix_initialize_virtual_pool_with_token2022.rs
- programs/dynamic-bonding-curve/src/instructions/migration/dynamic_amm_v2/damm_v2_utils.rs
- programs/dynamic-bonding-curve/src/instructions/migration/dynamic_amm_v2/migrate_damm_v2_initialize_pool.rs
- programs/dynamic-bonding-curve/src/instructions/migration/ix_withdraw_migration_fee.rs
- programs/dynamic-bonding-curve/src/instructions/migration/meteora_damm/migrate_meteora_damm_initialize_pool.rs
- programs/dynamic-bonding-curve/src/instructions/migration/withdraw_leftover.rs
- programs/dynamic-bonding-curve/src/instructions/operator/ix_claim_protocol_fee.rs
- programs/dynamic-bonding-curve/src/instructions/operator/ix_claim_protocol_pool_creation_fee.rs
- programs/dynamic-bonding-curve/src/instructions/operator/mod.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_claim_partner_pool_creation_fee.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_claim_partner_trading_fee.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_create_config.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_withdraw_partner_surplus.rs
- programs/dynamic-bonding-curve/src/instructions/swap/ix_swap.rs
- programs/dynamic-bonding-curve/src/instructions/swap/mod.rs
- programs/dynamic-bonding-curve/src/instructions/swap/swap_exact_in.rs
- programs/dynamic-bonding-curve/src/instructions/swap/swap_exact_out.rs
- programs/dynamic-bonding-curve/src/instructions/swap/swap_partial_fill.rs
- programs/dynamic-bonding-curve/src/lib.rs
- programs/dynamic-bonding-curve/src/params/liquidity_distribution.rs
- programs/dynamic-bonding-curve/src/state/config.rs
- programs/dynamic-bonding-curve/src/state/virtual_pool.rs
- programs/dynamic-bonding-curve/src/utis/token.rs

Findings

The security audit revealed:

- 0 critical issue
- 0 high issue



- 0 medium issue
- 0 low issue
- 2 informational issues

Further details, including the nature of these issues and recommendations for their remediation, are detailed in the subsequent sections of this report.



3 Summary of Findings

ID	Title	Severity	Status
01	Inconsistent Comment and Field Values	Informational	Fixed
02	Missing validation in pow	Informational	Fixed



4 Key Findings and Recommendations

4.1 Informational and Undetermined Issues

Inconsistent Comment and Field Values

Severity: Informational

Status: Fixed

Target: Smart Contract

Category: Code QA

The comment for `VirtualPool.migration_fee_withdraw_status` is as follows:

```
135    /// migration fee withdraw status
136    /// 0 is for partner,
137    /// 1 is for creator,
138    /// 2 is for protocol claim migration fee
139    pub migration_fee_withdraw_status: u8,
```

[programs/dynamic-bonding-curve/src/state/virtual_pool.rs#L135-L139](#)

However, the definitions of `partner` and `creator` use different values:

```
166    pub const PARTNER_MIGRATION_FEE_MASK: u8 = 0b100;
167    pub const CREATOR_MIGRATION_FEE_MASK: u8 = 0b010;
```

[programs/dynamic-bonding-curve/src/state/virtual_pool.rs#L166-L167](#)

The values used in the comment and in the implementation are inconsistent. It's recommended to update the comment to ensure it matches the implementation.

Missing validation in pow

Severity: Informational

Status: Fixed

Target: Smart Contract

Category: Data Validation

In the `pow` function, there is no check for the case where `exp == i32::MIN`. It's recommended to add a check similar to the one in DAMM v2:

```
37    // https://doc.rust-lang.org/std/primitive.i32.html#method.abs
38    // The absolute value of i32::MIN cannot be represented as an i32, an
39    if exp == i32::MIN {
40        return None;
41    }
```

[programs/cp-amm/src/math/fee_math.rs#L37-L41](#)



5 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

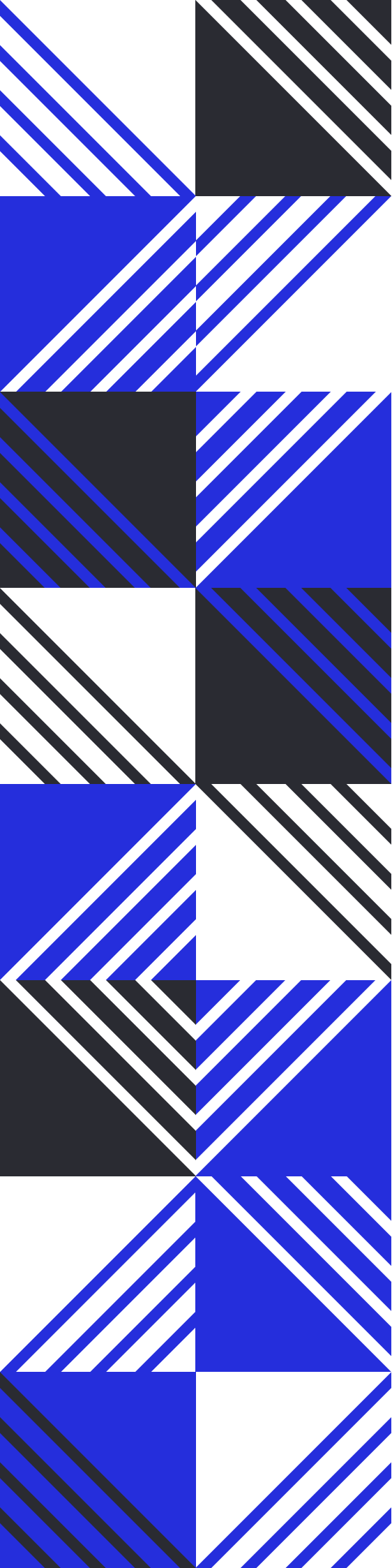
We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.



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